

KALIE KNECHT

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SUMMARY

Applied data scientist specializing in artificial intelligence, machine learning, and computer vision for nuclear nonproliferation and radiological search applications. Experienced in deep learning, reinforcement learning, and multimodal data fusion for real-world robotic and imaging systems. Strong record of interdisciplinary research, software development, and field deployment.

EDUCATION

University of California, Berkeley

Doctor of Philosophy in Nuclear Engineering
Graduate Certificate in Applied Data Science

Berkeley, CA

May 2024

December 2022

University of Tennessee

Bachelor of Science in Honors Nuclear Engineering

Knoxville, TN

May 2019

SELECTED RESEARCH EXPERIENCE

Oak Ridge National Laboratory

Applied Data Scientist

Oak Ridge, TN

September 2024 - Present

- Developing and deploying ML pipelines for autonomous radiation mapping, integrating real-time multimodal sensor data (e.g., camera, NaI, LiDAR) with custom exploration and proximity-based decision frameworks on a quadruped robotic platform.
- Training and evaluating deep learning models for anomaly detection and spectral classification, including transformer-based models and autoencoders applied to simulated spectro-temporal radiation data and KUT source mixtures.
- Designing and executing large-scale prompt engineering and evaluation pipelines for large language model (LLM)-based question generation, benchmarking retrieval-augmented generation (RAG) and non-RAG agents using Qwen, Gemma, and DeepSeek models for domain-specific datasets.
- Developing ML-driven tools for sensor anomaly detection and vehicle re-identification and tracking using object detection models and LLMs; integrated outputs into a real-time data state-of-health framework for radiation sensing nodes across field deployments.

Lawrence Berkeley National Laboratory

Graduate Student Researcher

Berkeley, CA

August 2019 - May 2024

- Used scene data collected by a free-moving gamma-ray detector and contextual sensor package (LiDAR, IMU, Depth Camera) to optimize measurement positions for quantitative gamma-ray imaging.
- Used 3D instance and semantic segmentation machine learning techniques to identify and label discrete objects in RGB-D and LiDAR point clouds.
- Developed a genetic algorithm in Python to perform multi-objective optimization of measurement positions for quantitative gamma-ray imaging.

Los Alamos National Laboratory

Space Science and Applications Intern

Los Alamos, NM

June 2020 - August 2020

- Developed software in Python to analyze the charge collection in a two-pixel semiconductor detector to be used in a space radiation telescope.

TEACHING EXPERIENCE

University of California, Berkeley

Graduate Student Instructor

Berkeley, CA

Fall 2020 and Fall 2021

- Undergraduate radiation detection (NE 104): semiconductor and scintillator detector operation, manufacturing, signal generation, readout techniques, applications and limitations.
- Supervised students in the laboratory and instructed students in scientific writing.
- Recorded laboratory experiments and edited videos to ensure safe & equitable learning during the COVID-19 pandemic.

University of Tennessee

Undergraduate Teaching Assistant

Knoxville, TN

August 2018 - May 2019

- Developed weekly review sessions for Thermal Science and Reactor Theory courses.
- Provided tutoring services for students enrolled in Thermal Science and Reactor Theory.

SKILLS

Programming Languages:	Python, C++, OpenCL, Fortran
Machine Learning & AI Techniques:	Deep Learning, Reinforcement Learning, Autoencoders, Triplet Loss, Anomaly Detection, Embedding Analysis, Transformers, LLMs, Prompt Engineering, RAG
Libraries & Frameworks:	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
Computer Vision:	Instance Segmentation, Object Detection, Semantic Segmentation, SLAM, OpenCV, 3D Perception, Point Cloud Processing, Sensor Fusion
Data Engineering:	Pandas, NumPy, SciPy, SQL, HDF5, ROS, PCAP
Scientific Computing:	MCNP, GOTHIC, Cyclus, Radiation Transport Modeling
Version Control:	Git, SVN
Containers & Virtualization:	Docker, Conda
Markup & Documentation:	LaTeX, Markdown, Sphinx
Operating Systems:	Linux, Windows, macOS
Laboratory Skills:	Radiation Measurements, Gamma-Ray Spectroscopy, X-ray Diffraction

LEADERSHIP EXPERIENCE

Society of Women Engineers

Smoky Mountain SWE President

June 2026 - Present

- Managing a team of 8 officers.
- Maintaining website and public calendar.

Smoky Mountain SWE President-Elect

June 2025 - May 2026

- Organizing fundraising events at local restaurants and developing merchandise to raise funds for section operations.
- Volunteering at outreach events with the UTK SWE section to encourage K-12 girls to pursue careers in STEM through hands-on engineering activities.
- Mentoring 2 undergraduate engineering students from UTK SWE through their alumni mentoring program to support their academic and professional development through regular meetings.

TECHNICAL MENTORING

Ellyott Siffring, Undergraduate Student	Summer 2026
<i>"LAMP Radiation and Mapping Validation"</i>	
Lance Drouet, Graduate Student	Summer 2025 - Present
<i>"Anomaly detection algorithms for nuclear source search"</i>	
Peyton Andras, Graduate Student	Summer 2025, 2026
<i>"Tracking vehicles with a distributed tire-pressure monitoring signal sensor network"</i>	
<i>"NeuroStereo: A Calibrated Event-Based Stereo Vision Pipeline for Dynamic Roadside Depth Estimation"</i>	Summer 2025
	Summer 2026

PROFESSIONAL SERVICE

2025 IEEE SORMA West Topic Convener
 2025 IEEE SORMA West Session Chair
 Reviewed 2 manuscripts for IEEE TNS in 2025, 2 in 2026
 2023 IEEE Nuclear Science Symposium Reviewer

HONORS AND AWARDS

Oak Ridge National Laboratory

Supplemental Performance Award for fieldwork efforts June 2026
 Supplemental Performance Award for mentorship September 2025

UC Berkeley Department of Nuclear Engineering

Virgil Schrock Award for Outstanding Service May 2022
 Best Student Paper on Radiation Detection and Imaging Dec 2021
"Polaris-LAMP: Multi-Modal 3-D Image Reconstruction With a Commercial Gamma-Ray Imager"
 Virgil Schrock Award for Outstanding Service May 2021
 Virgil Schrock Award for Outstanding Mentorship May 2020

FIRST AUTHOR PUBLICATIONS, PROCEEDINGS, & PAPERS

1. **K. Knecht et al.**, "Integrating Agentic AI and Real-Time Radiation Mapping for Autonomous Source Search," *in progress*.
2. **K. Knecht et al.**, "Coordinated Autonomous Navigable Instruments for Nuclear Emergency Search (CANINES)," in Proc. INMM Annual Meeting, Washington, D.C., 2025.
3. **K. Knecht**, "Enhanced use of contextual data for quantitative gamma-ray imaging in nuclear safeguards applications," Dissertation No. 31297598, University of California, Berkeley, 2024. Available: <https://escholarship.org/uc/item/3627r1jx>.
4. **K. Knecht et al.**, "Enhanced use of contextual data for quantitative Compton imaging," in Proc. IEEE NSS/MIC/RTSD, Vancouver, BC, Canada, 2023, pp. 1-1, doi: 10.1109/NSSMICRTSD49126.2023.10337904.
5. **K. Knecht et al.**, "Scene-informed Optimal Measurement Positions for Quantitative Safeguards Measurements," in Proc. INMM and ESARDA Joint Annual Meeting, Vienna, Austria, 2023, Available: <https://resources.inmm.org/annual-meeting-proceedings/scene-informed-optimal-measurement-positions-quantitative-safeguards>.
6. **K. Knecht**, "From the Field," in Berkeley Science Review, May 2022, Available: <https://www.berkeleysciencereview.com/article/2022/05/03/from-the-field>.
7. **K. Knecht et al.**, "3D Compton Imaging of Distributed Sources around the Chernobyl Nuclear Power Plant," in Proc. IEEE Nuclear Science Symposium and Medical Imaging Conference (NSS/MIC), Piscataway, NJ, 2021, pp. 1-4, doi: 10.1109/NSS/MIC44867.2021.9875432.
8. J. Hecla and **K. Knecht et al.**, "Polaris-LAMP: Multi-modal 3-D Image Reconstruction with a Commercial Gamma-ray Imager," in IEEE TNS, vol. 68, no. 10, pp. 2539-2549, Oct. 2021, doi: 10.1109/TNS.2021.3110162.
9. **K. Knecht et al.**, "Evaluating 3D gamma-ray Imaging Techniques for Distributed Sources at the Fukushima Daiichi Nuclear Power Station," in Proc. IEEE NSS/MIC, Boston, MA 2020, pp. 1-5, doi: 10.1109/NSS/MIC42677.2020.9507840.

ORAL PRESENTATIONS

1. **K. Knecht et al.**, "Multi-Agent Autonomy for Radioactive Mapping in Complex Environments," at INMM Annual Meeting, Austin, TX, 2026.
2. **K. Knecht et al.**, "Multimodal Sensor Networks and AI-Driven Data Fusion for Radiological Detection," at INMM AI4NSSN Workshop, Albuquerque, NM, 2026.
3. **K. Knecht et al.**, "Coordinated Autonomous Navigable Instruments for Nuclear Emergency Search (CANINES)," at INMM Annual Meeting, Washington, D.C., 2025.
4. **K. Knecht et al.**, "Integrating Agentic AI and Real-Time Radiation Mapping for Autonomous Source Search," at SORMA West, Berkeley, CA, 2025.
5. **K. Knecht et al.**, "Enhanced Use of Contextual Data for Quantitative Compton Imaging," at IEEE NSS/MIC/RTSD, Vancouver, BC, Canada, 2023.
6. **K. Knecht et al.**, "Scene-informed Optimal Measurement Positions for Quantitative Safeguards Measurements," at INMM and ESARDA Joint Annual Meeting, Vienna, Austria, 2023.
7. **K. Knecht**, "Scene-informed Optimal Measurement Positions for Quantitative Safeguards Measurements," at University Program Review (UPR), Berkeley, CA, 2023.
8. **K. Knecht et al.**, "Scene-informed Optimization of Measurement Locations for Radiological Assessments," at IEEE NSS/MIC, Milan, Italy, 2022.
9. **K. Knecht**, "Scene-informed Optimization of Measurement Locations for Radiological Assessments," at UPR, Ann Arbor, MI, 2022.
10. **K. Knecht et al.**, "3D Compton Imaging of Distributed Sources around the Chernobyl Nuclear Power Plant," at IEEE NSS/MIC, Piscataway, NJ, 2021.
11. **K. Knecht**, "3D Compton Imaging of Distributed Sources Around the Chernobyl NPP," at UPR, 2021.

12. **K. Knecht** *et al.*, "Evaluating 3D Gamma-ray Imaging Techniques for Distributed Sources at the Fukushima Daiichi Nuclear Power Station," at IEEE NSS/MIC, Boston, MA, 2020.
13. **K. Knecht**, C. Roecker, and K. Smith, "Signal Generation in CdTe Detector with an Active Guard Ring," at LANL Keepin Program Student Presentation Session, 2020.

INVITED TALKS

1. **K. Knecht**, "AI for Source Search," at DNN TeCH Modern and Emerging AI Summer School, Oak Ridge, TN, 2026.
2. **K. Knecht**, "Scene Data Fusion Overview" at PNNL Radiation Detection for Nuclear Security Summer School, Richland, WA, 2025.

POSTER PRESENTATIONS

1. **K. Knecht** *et al.*, "Multi-Agent Autonomy for Radioactive Mapping in Complex Environments," at INMM Annual Meeting, Austin, TX, 2026.
2. **K. Knecht**, *et al.*, "Coordinated Autonomous Navigable Instruments for Nuclear Emergency Search (CANINES)", at ORNL AI Expo, Oak Ridge, TN, 2025.
3. **K. Knecht**, *et al.*, "Scene-informed Optimization of Measurement Locations for Radiological Assessment," at UPR, Berkeley, CA, 2023.
4. **K. Knecht**, *et al.*, "3D Radiological Mapping in Chernobyl," at Nuclear Science & Security Consortium Fall Workshop and Advisory Board Meeting, 2022.
5. **K. Knecht**, *et al.*, "3D Radiological Mapping in Chernobyl," at Lawrence Berkeley National Laboratory Nuclear Science Division Director's Review, 2021.
6. **K. Knecht**, *et al.*, "Improving Facility-Specific Safeguards with Data Analytics," at ORNL Summer Student Poster Presentations, Oak Ridge, TN, 2019.
7. **K. Knecht**, *et al.*, "Solid State Synthesis of $\text{Nd}_2\text{Zr}_2\text{O}_7$ and Study of its Thermal Properties Using *in-situ* X-ray Diffraction," at The Minerals, Metals and Materials Annual Meeting & Exhibition, Nashville, TN, 2016.

OTHER PUBLICATIONS, PROCEEDINGS, & PAPERS

1. S. Ghanem, **K. Knecht**, R. Kereks, B. Longmire, D. Peplow, A. Rowe, J. Sparger, E. Wong, and D. Archer, "Compact Multimodal Sensor Networks for Radiological Detection and Vehicle Reidentification", *in progress*.
2. D. Gunter, R. Barnowski, A. Haefner, D. Hellfeld, **K. Knecht**, M. Bandstra, V. Negut, E. Frame, R. Pavlovsky, T. H. Y. Joshi, J. Vavrek, B. Quiter, and K. Vetter. "Modeling the Compton Camera Response for Extended Voxel Sources." Lawrence Berkeley National Laboratory Report #2001559, 2024.
3. K.-P. Ziock, F. Gonzalez, J. Daughetee, A. Laminack, K. Schmitt, R. Cooper, **K. Knecht**, D. Goodman, and W. Kaye, "2023 Annual Report: Multimodal Data Fusion with 3D Gamma-Ray Imaging for Safeguards," ORNL SPR 2023 3216, 2023, Available: <https://www.osti.gov/biblio/2439886>.
4. D. Hellfeld, M. Folsom, T. H. Y. Joshi, **K. Knecht**, J. Lee, D. Gunter, K. Schmitt, J. Daughetee, K.-P. Ziock, S. Horne, S. Brown, D. Goodman, "Quantitative Compton Imaging in 3D," in Proc. INMM Annual Meeting, 2022, Available: <https://resources.inmm.org/annual-meeting-proceedings/quantitative-compton-imaging-3d>.
5. J. Hecla, **K. Knecht**, K. Vetter, T. H. Y. Joshi, A. Haefner, and R. Pavlovsky. "Three-dimensional Radiation Mapping at Chernobyl Nuclear Power Plant," in Proc. Sixth International Conference on Nuclear Decommissioning and Environment Recovery, Ukraine, 2021, pp. 15.
6. M. Hines, M. Underwood, A. Naylor, M. Burrell, L. Clowers, **K. Knecht**, and G.I. Maldonado. "Equilibrium core design of a NuScale designed small modular reactor using CASL's virtual environment for reactor applications (VERA)," in Transactions of the American Nuclear Society, Vol. 121, Washington, D.C., November 2019.

OTHER PRESENTATIONS

1. H. S. Parilla, M. S. Bandstra, R. J. Cooper, C. Jones, **K. Knecht**, A. Nicholson, and J. Sparger, "Linear and interpretable spectro-temporal background templates for gamma-ray anomaly detection" at SORMA West, Berkeley, CA, 2025.